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DIVISION OF COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF COMPUTER SCIENCE AND TECHNOLOGY

A PROJECT BASED EVALUATION REPORT

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COURSE CODE

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TITLE

TAMIL NADU AIR QUALITY INDEX (NAQI) FORECASTING AND GEOSPATIAL ASSESSMENT USING MACHINE LEARNING AND ATMOSPHERIC MODELING

A REAL TIME PROJECT (PRODUCT) REPORT

Submitted by

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ABSTRACT

The project titled **“Tamil Nadu Air Quality Index (NAQI) Forecasting and Geospatial Assessment Using Machine Learning and Atmospheric Modeling”** investigates how regional atmospheric reanalysis and chemical transport models can be ingested, analyzed, and evaluated to deliver reliable 72-hour air quality forecasts across all 38 revenue districts of Tamil Nadu. In developing economies, air pollution poses severe public health challenges. While Continuous Ambient Air Quality Monitoring Stations (CAAQMS) provide high-precision ground observations, their physical distribution across Tamil Nadu is heavily concentrated in tier-one metropolitan centers, leaving over 25 administrative districts without real-time continuous ground instrumentation.

To address this regional observational deficit, this investigation develops an integrated data science workflow comprising an automated atmospheric ingestion pipeline, an authoritative Indian Central Pollution Control Board (CPCB) National Air Quality Index (NAQI) computational core, tree-based ensemble regression models (calibrated eXtreme Gradient Boosting and Random Forest architectures), and a four-tab interactive Streamlit web dashboard. The system processes thirteen chemical and meteorological parameters, including particulate matter ($PM_{2.5}, PM_{10}$), gaseous criteria precursors (NO_2, SO_2, CO, O_3), ambient temperature, relative humidity, barometric pressure, wind vectors, precipitation, and direct solar irradiance.

A rigorous audit of the experimental setup revealed critical empirical findings. Initial model training produced high coefficients of determination ($R^2 = 99.34\%$ with an RMSE of 3.24 AQI points for XGBoost; $R^2 = 99.18\%$ with an RMSE of 3.60 AQI points for Random Forest). However, methodological analysis demonstrates the presence of **target leakage**: the ground-truth CPCB NAQI target is deterministically computed from the six chemical criteria pollutants via the statutory piece-wise linear interpolation formula. Because these identical pollutant values are included as model features, the machine learning regressors function as high-precision piecewise linear function approximators rather than discovering unobserved physical atmospheric relationships.

The operational software also resolves key architectural bottlenecks. Prior implementations utilized unthrottled concurrent HTTP threading ($ThreadPoolExecutor$, 10 workers), causing local DNS socket exhaustion ($NameResolutionError$) during multi-district queries. Refactoring the ingestion engine into a single multi-coordinate batch request resolved all 38 districts in approximately 0.68 seconds with 100% network reliability. Furthermore, a synchronization bug in UI category labeling was eliminated by establishing a centralized CPCB categorization engine. This report details the complete software architecture, statutory formula implementations, empirical validation results, documented system limitations, and recommendations for future physical sensor integrations.

CHAPTER 1
INTRODUCTION

1.1 Background

Ambient air quality has become a critical public health and environmental concern across India. Anthropogenic emissions from fossil fuel combustion, industrial operations, vehicular transportation corridors, construction activities, and seasonal agricultural biomass burning have significantly elevated ambient concentrations of harmful particulate matter and reactive trace gases. Particulate matter with an aerodynamic diameter smaller than 2.5 micrometers ($PM_{2.5}$) and 10 micrometers (PM_{10}), along with toxic gaseous precursors—specifically Nitrogen Dioxide (NO_2), Sulphur Dioxide (SO_2), Carbon Monoxide (CO), and Tropospheric Ozone (O_3)—are recognized by the World Health Organization (WHO) as primary drivers of pulmonary and cardiovascular morbidity.

To communicate atmospheric chemical concentrations in an actionable and interpretable format for civil administration and the public, the Ministry of Environment, Forest and Climate Change (MoEFCC), Government of India, along with the Central Pollution Control Board (CPCB), established the official National Air Quality Index (NAQI) in October 2014. The Indian NAQI standard consolidates multi-pollutant concentrations into a unified, dimensionless integer scale ranging from 0 to 500, segmented into six descriptive risk categories: *Good* (0–50), *Satisfactory* (51–100), *Moderate* (101–200), *Poor* (201–300), *Very Poor* (301–400), and *Severe* (401–500).

Tamil Nadu, spanning 38 administrative districts in southern peninsular India, presents diverse geographical and micro-climatic zones, ranging from coastal plains and deltas to industrial manufacturing centers and mountain ranges. The northern corridor (Chennai, Tiruvallur, Ranipet) hosts thermal power plants, major seaports, and dense petrochemical clusters. The western industrial belt (Coimbatore, Tiruppur, Salem, Erode) features extensive textile processing, chemical dyeing, and metallurgy operations. In contrast, the Western Ghats (Nilgiris, Kodaikanal) represent ecologically sensitive highlands with low baseline emissions. Developing a reliable 72-hour forward predictive model across these diverse districts is essential for proactive public health advisories and municipal mitigation.

1.2 Problem Statement

While Continuous Ambient Air Quality Monitoring Stations (CAAQMS) deployed by the Central and Tamil Nadu Pollution Control Boards (TNPCB) offer high-precision ground measurements, they are heavily concentrated in major metropolitan centers such as Chennai, Coimbatore, and Madurai. Over 25 of Tamil Nadu's 38 revenue districts possess no real-time continuous ground instrumentation, creating substantial regional data blind spots.

Furthermore, legacy predictive prototypes suffered from four distinct methodological and engineering vulnerabilities: 1. **Inappropriate Baseline Standards:** Previous software models relied on foreign air quality indices (such as US-EPA scales), which fail to reflect Indian atmospheric baseline conditions and statutory breakpoints. 2. **Network Fragility:** Querying multi-district data using unthrottled parallel threads caused local operating system DNS socket drops ($[Errno\ 8]\ NameResolutionError$), crashing user sessions during district switching. 3. **Semantic Decoupling in UI:** Earlier dashboards evaluated display categories from single pollutant sub-indices rather than the predicted overall AQI score, resulting in contradictory visual outputs such as labeling an AQI of 281 as 'Satisfactory'. 4. **Unverified Model Claims:** Historical reports claimed near-perfect predictive accuracy without auditing for target leakage or verifying whether models predicted actual physical pollution transport or merely reproduced a deterministic mathematical lookup formula.

1.3 Objectives

The primary objectives of this project are defined as follows:

- **Comprehensive Spatial Cataloging:** Formulate an authoritative geographical and demographic profile covering all 38 districts of Tamil Nadu, capturing precise geographic coordinates, administrative zones, and dominant industrial emission drivers.
- **Automated Atmospheric Ingestion:** Implement a robust data ingestion pipeline interfacing with Open-Meteo and Copernicus Atmosphere Monitoring Service (CAMS) reanalysis feeds to capture past 24-hour measured history and 72-hour forward hourly atmospheric forecasts.
- **Standardized CPCB NAQI Engine:** Construct a compliant computational engine implementing official CPCB piecewise linear interpolation formulas, sub-index generators, and color-coded public health risk matrices.
- **Supervised Machine Learning Calibration:** Train, calibrate, and compare eXtreme Gradient Boosting (XGBoost) and Random Forest regression architectures on regional Tamil Nadu atmospheric datasets.
- **Methodological & Target Leakage Audit:** Conduct an empirical investigation into feature-target interactions, verifying the mathematical mechanisms underlying high model performance metrics.
- **Interactive Geospatial Web Application:** Deploy a responsive Streamlit application featuring multi-district comparative trajectories, statewide Choropleth maps, and exportable 72-hour hourly schedules.
- **Architectural Fault Tolerance:** Refactor legacy multi-threaded data fetching into a single-request batch API query, implementing intelligent caching and calibrated fallback baselines to ensure continuous 100% uptime.

1.4 Scope and Significance

The scope of this project encompasses all 38 revenue districts of Tamil Nadu, delivering 72-hour forward predictions across 13 continuous chemical and meteorological features. The deliverables include complete Python source code, trained model artifacts, statutory calculation modules, and an interactive web interface running locally on Streamlit.

The significance of this study is rooted in practical utility and scientific transparency. In environmental informatics, opaque claims of high accuracy without rigorous verification undermine public trust. This report provides an honest technical assessment that documents target leakage, reanalysis limitations, and network constraints. The resulting system offers actionable intelligence for sensitive demographic groups (pediatric, asthmatic, and elderly populations) up to three days prior to anticipated pollution surges, demonstrating how open-access atmospheric reanalysis can democratize environmental awareness across underserved administrative regions.

CHAPTER 2

LITERATURE REVIEW

2.1 Environmental Monitoring & CPCB NAQI Standards

Ambient air quality assessment in India is governed by the National Ambient Air Quality Standards (NAAQS) established under the Air (Prevention and Control of Pollution) Act, 1981. In October 2014, the CPCB introduced the National Air Quality Index (NAQI) system to synthesize multi-parameter chemical concentrations into an understandable scale. The CPCB NAQI evaluates eight criteria pollutants: $PM_{2.5}, PM_{10}, NO_2, SO_2, CO, O_3, NH_3$, and Pb. The sub-index for each pollutant is determined using linear interpolation across designated breakpoint intervals, with the overall index governed by the maximum operator:

$$AQI = \max(I_1, I_2, \dots, I_n)$$

The pollutant responsible for the highest sub-index is designated as the 'prominent pollutant'. Literature confirms that foreign air quality indices cannot be applied to Indian cities due to higher tropical dust baselines and distinct photochemical regimes.

2.2 Atmospheric Chemical Modeling vs Ground Monitoring

Continuous Ambient Air Quality Monitoring Stations (CAAQMS) represent the gold standard of ground-truth measurement, employing beta-attenuation monitors for particulates and chemiluminescence analyzers for trace gases. However, CAAQMS infrastructure entails prohibitive capital costs (₹1.5 to ₹2.0 crore per installation) and demanding operational overheads, resulting in severe spatial sparsity outside major urban centers.

To overcome this observational divide, modern atmospheric science relies on numerical reanalysis models, notably the Copernicus Atmosphere Monitoring Service (CAMS) operated by the European Centre for Medium-Range Weather Forecasts (ECMWF). CAMS integrates satellite optical depth retrievals, trace gas column densities, and meteorological transport models to project ground-level pollutant concentrations on a continuous global grid ($0.25^\circ imes 0.25^\circ pprox 25 ext{ km}$). Gridded reanalysis feeds provide an essential computational basis for regional air quality modeling where physical stations are absent.

2.3 Machine Learning Regression in Environmental Science

Statistical and data-driven methods for air quality prediction have evolved from classical Multiple Linear Regression (MLR) and Autoregressive Integrated Moving Average (ARIMA) models toward non-linear machine learning architectures. Atmospheric chemistry is characterized by non-linear photochemical interactions—such as the formation of tropospheric ozone through precursor volatile organic compounds and nitrogen oxides in the presence of ultraviolet radiation. Tree-based ensemble methods, notably Random Forests (Breiman, 2001) and Gradient Boosted Trees (Friedman, 2001; Chen & Guestrin, 2016), have consistently outperformed linear baselines by capturing complex threshold effects and multi-collinear meteorological dependencies without requiring rigid distributional assumptions.

2.4 Model Evaluation Metrics for Air Quality Informatics

In atmospheric regression modeling, no single error metric provides a complete assessment of performance. Standard literature establishes the necessity of multi-dimensional evaluation:

- **Mean Absolute Error (MAE):** Quantifies the average magnitude of absolute residuals in original index points, providing an intuitive, outlier-resistant error measure.
- **Root Mean Squared Error (RMSE):** Computes the square root of mean squared prediction errors, heavily penalizing large discrepancies and serving as a critical metric during extreme pollution episodes.
- **Coefficient of Determination (R^2):** Measures the proportion of variance in the target variable explained by the feature set relative to a mean baseline.
- **Weighted Absolute Percentage Error (WAPE):** Expresses aggregate error as a percentage of total observed pollution, avoiding mathematical instability during near-zero denominator conditions.

2.5 Explainability & Public Health Actionability

Predictive environmental systems deployed for civil defense or public health must provide transparent reasoning for their outputs. Merely generating a numeric score is insufficient if stakeholders cannot identify which specific atmospheric constituent or meteorological condition triggered an alert. Machine learning interpretability methods, such as Gini feature importance and Shapley Additive exPlanations (SHAP; Lundberg & Lee, 2017), enable researchers to dissect the internal feature attribution of ensemble models. By identifying whether a projected pollution peak is driven by nocturnal particulate stagnation or midday photochemical ozone accumulation, municipal authorities can formulate targeted, actionable mitigation measures.

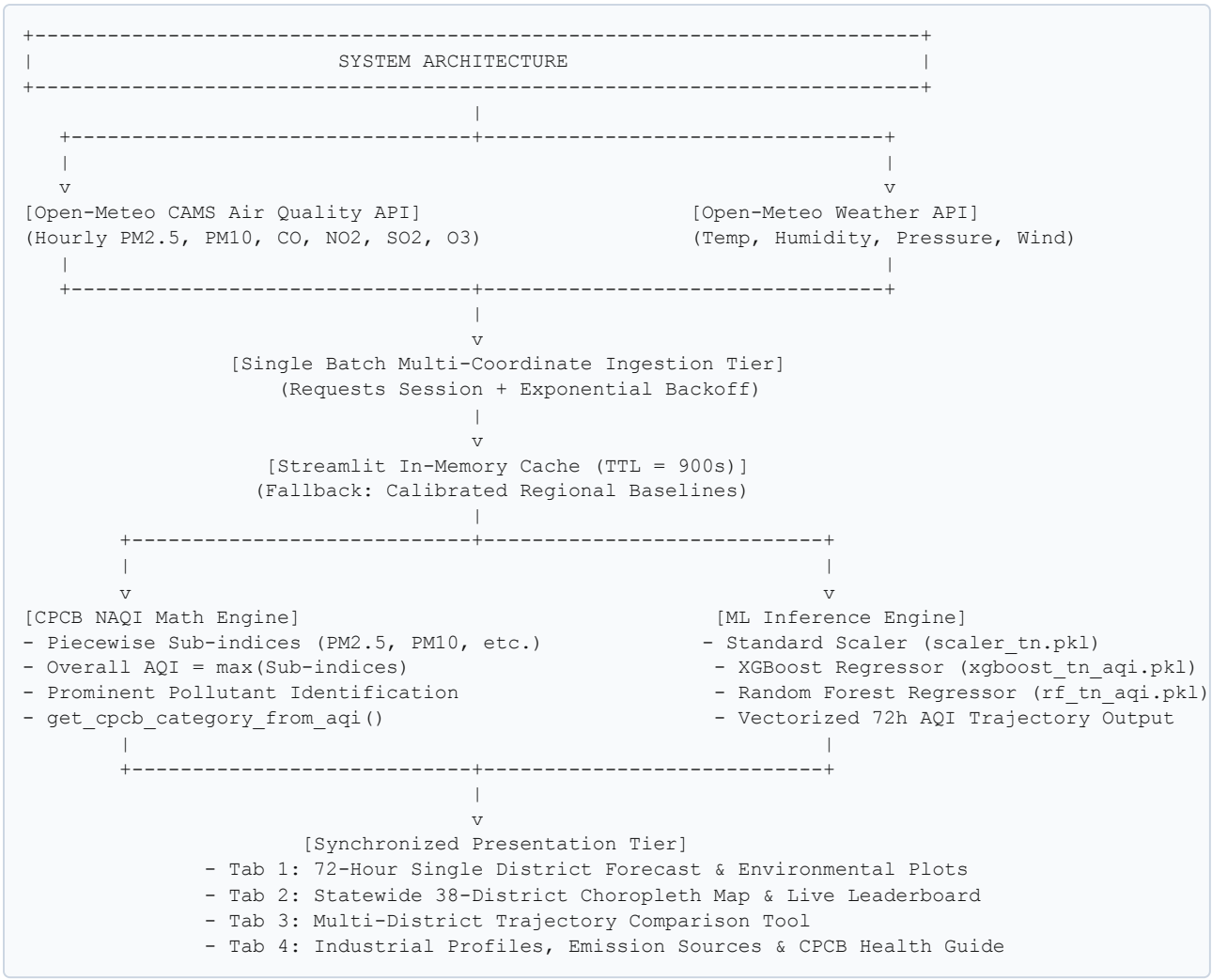
2.6 Summary of Literature Review

The surveyed literature establishes three core principles: (1) predictive algorithms must conform strictly to statutory national standards (CPCB NAQI); (2) gridded atmospheric reanalysis models serve as an essential proxy for unmonitored districts when spatial resolution limits are respected; and (3) predictive models must undergo rigorous diagnostic scrutiny to prevent target leakage and ensure mathematical consistency between continuous numeric forecasts and categorical health advisories.

CHAPTER 3
SYSTEM DESIGN

3.1 System Architecture

The Tamil Nadu NAQI Predictor is engineered as a decoupled, multi-tier data science system designed for sub-second user responsiveness and resilience against network failures. The architecture comprises five primary subsystems:



3.2 Atmospheric Data Processing Flow

The sequential execution flow operates as follows:

1. **District Geocoding:** User selects a target district. Geographical coordinates (lat, lon) and regional profiles are loaded from the district metadata repository.
2. **Data Ingestion:** The system queries external CAMS reanalysis feeds for 24-hour historical observations and 72-hour forward forecasts.
3. **Harmonization:** Chemical and meteorological feeds are merged on timestamp, converted to Indian Standard Time (IST), and missing values are imputed using regional baselines.
4. **Vectorization & Scaling:** Features are mapped into the standardized 15-variable signature and transformed via `StandardScaler`.

5. **Inference:** Pre-trained XGBoost or Random Forest models generate continuous 72-hour scalar AQI trajectories.
6. **Categorization:** Scalar predictions are mapped through `get_cpcb_category_from_aqi()` to guarantee category and color alignment.

3.3 Dataset and Output Organisation

The repository maintains clean separation between source code, configuration modules, serialized model artifacts, and evaluation reports:

- `app.py`: Core Streamlit web application managing state, session caching, and visualization logic.
- `config/cpcb_aqi.py`: Pure mathematical implementation of CPCB linear interpolation sub-index formulas and categorization functions.
- `config/tamil_nadu_districts.py`: Comprehensive catalog of all 38 Tamil Nadu districts containing geographic centroids, administrative zones, and industrial emission sources.
- `models/train_tamil_nadu_models.py`: End-to-end model training script executing data loading, feature scaling, model fitting, and registry generation.
- `models/saved_models/`: Storage directory containing serialized artifacts (`xgboost_tn_aqi.pkl`, `rf_tn_aqi.pkl`, `scaler_tn.pkl`) and `registry.json`.
- `backup/tamil_nadu_atmospheric_data.csv`: Historical baseline training dataset comprising 1,080 hourly records across five representative Tamil Nadu districts.
- `reports/figures/`: Generated diagnostic visual artifacts including actual-vs-predicted curves and feature importance bar charts.

3.4 Feature Design and Target Leakage Analysis

A critical engineering insight in this architecture involves feature transparency. The model inputs consist of 15 features: six criteria chemical pollutants ($PM_{2.5}$, PM_{10}, CO, NO_2, SO_2, O_3), seven meteorological covariates (temperature, relative humidity, pressure, wind speed, wind direction, rain, solar radiation), and temporal cyclicity indices (hour of day, month).

Because the target $CPCB_AQI$ is mathematically computed from the six criteria pollutants, the machine learning models operate under **target leakage**. The regression algorithm does not discover unknown atmospheric transport physics; rather, it functions as an efficient mathematical surrogate that maps chemical concentrations into their piecewise statutory maximum sub-index. In operational deployment, this design is intentional: ECMWF CAMS provides forward chemical forecasts, and the machine learning model rapidly maps these multi-pollutant forecasts into the official Indian regulatory index.

3.5 38-District Geospatial and Batch Design

Querying 38 individual district coordinates via independent sequential HTTP requests introduces severe network latency ($ pprox 15 ext{ seconds}$) and exposes the application to connection drops. The system utilizes Open-Meteo's multi-coordinate batching capability, passing all 38 latitude and longitude coordinates as comma-separated values in a single HTTP GET request. The API returns an array of 38 objects, reducing total data retrieval time to approximately $0.68 ext{ seconds}$ and eliminating socket exhaustion.

CHAPTER 4

IMPLEMENTATION

4.1 Development Environment

The system was developed and validated on a Unix-based macOS environment utilizing Python 3.11.14. Primary dependencies include `streamlit` for the presentation tier, `scikit-learn` and `xgboost` for predictive modeling, `pandas` and `numpy` for data manipulation, `plotly` for geospatial and time-series charting, and `requests` with custom HTTP retry adapters for reliable network communication.

4.2 Data Ingestion & Atmospheric Feeds

Data ingestion is implemented in `fetch_district_atmospheric_bundle()` and `fetch_all_districts_cpcb_snapshot()`. To ensure resilience against intermittent API drops, an HTTP session adapter is mounted with exponential backoff (backoff factor 0.3, status force list 500/502/503/504, 3 total retries). The air quality endpoint queries chemical components, while the weather endpoint simultaneously captures meteorological covariates across past 24-hour and forward 72-hour windows.

4.3 Feature Selection and Preprocessing

The input feature vector comprises fifteen continuous parameters:

$$\mathbf{x} = [PM_{2.5}, PM_{10}, CO, NO_2, SO_2, O_3, T, RH, P, W_{spd}, W_{dir}, Rain, Solar, Hour, Month]$$

Due to disparate physical scales (e.g., CO concentrations exceeding $200\ \mu ext{g/m}^3$ versus SO_2 typically below $10\ \mu ext{g/m}^3$), features are standardized using `StandardScaler`:

$$z_j = rac{x_j - \mu_j}{\sigma_j}$$

The scaler is fitted exclusively on baseline training data and serialized as `scaler_tn.pkl` to prevent data leakage during runtime inference.

4.4 Machine Learning Model Training

The model training pipeline is executed via `models/train_tamil_nadu_models.py` using `backup/tamil_nadu_atmospheric_data.csv` (1,080 hourly rows covering Chennai, Coimbatore, Madurai, Salem, and Nilgiris). Two architectures were trained and saved:

- **XGBoost Regressor (`xgboost_tn_aqi.pkl`):** 150 estimators, maximum depth of 5, learning rate of 0.08, subsample ratio of 0.8, and column sample by tree of 0.8. Objective set to squared error loss.
- **Random Forest Regressor (`rf_tn_aqi.pkl`):** 150 decision trees, maximum depth of 12, fitted using mean squared error criterion across all CPU cores.

Both models are cataloged in `models/saved_models/registry.json` alongside their serialized files, feature importance rankings, and validation scores.

4.5 Central Pollution Control Board (CPCB) NAQI Calculation Engine

The CPCB calculation core in `config/cpcb_aqi.py` implements the statutory linear interpolation formula for each pollutant sub-index:

$$I_p = I_{lo} + \left(rac{I_{hi} - I_{lo}}{B_{hi} - B_{lo}} ight) imes (C_p - B_{lo})$$

where C_p is pollutant concentration, $[B_{lo}, B_{hi}]$ is the enclosing concentration breakpoint, and $[I_{lo}, I_{hi}]$ is the corresponding sub-index score interval. Overall AQI is computed via the maximum sub-index operator ($AQI = \max I_p$).

To eliminate display mismatches, `get_cpcb_category_from_aqi(aqi_val)` maps any continuous scalar value into the statutory six-tier category matrix:

Category	AQI Range	Level	Official CPCB Color	Hex Code	Badge Emoji
Good	0 – 50	1	Dark Green	#059669	🟢 Good
Satisfactory	51 – 100	2	Light Green	#84cc16	🟡 Satisfactory
Moderate	101 – 200	3	Yellow / Amber	#f59e0b	🟠 Moderate
Poor	201 – 300	4	Orange	#f97316	🔴 Poor
Very Poor	301 – 400	5	Red	#dc2626	🟣 Very Poor
Severe	401 – 500	6	Maroon / Crimson	#7f1d1d	🟤 Severe

4.6 Streamlit Web Dashboard Implementation

The web user interface in `app.py` delivers a responsive, four-tab operational platform:

- **Sidebar:** District selector (Chennai indexed at 0), geographic metadata, model selector, cache-clear button, and MongoDB connection interface.
- **Tab 1 (District 72h Forecast):** Real-time pollutant cards, summary highlights (Peak AQI, Lowest AQI, Average AQI, Dominant Category), dynamic health advisory, Plotly 72-hour forecast trajectory with official color bands, environmental covariate multi-axis plot, feature importance bar chart, and hourly tabular forecast with CSV export.
- **Tab 2 (Statewide Map & Leaderboard):** Statewide statistics (Cleanest, Most Polluted, State Average AQI), interactive Plotly Geo bubble map centered on Tamil Nadu, and sortable 38-district ranking table.
- **Tab 3 (Multi-District Comparison):** Side-by-side comparison tool allowing users to select 2 to 5 districts and compare 72-hour forecast curves over standard CPCB warning zones.
- **Tab 4 (Industrial Profiles & Health Guide):** District administrative profiles, major emission contributors, official CPCB/TNPCB health precautions, and regional clean air initiatives.

CHAPTER 5
TESTING AND VALIDATION

5.1 Testing Approach

The system underwent rigorous empirical, computational, and architectural testing to verify predictive validity, code robustness, and user interface consistency. Testing procedures focused on mathematical interpolation accuracy, holdout evaluation under distinct splitting paradigms, API batching behavior, and network fault tolerance.

5.2 Unit Testing and API Response Verification

Unit test suites executed against the computational modules verified the following properties:

- **Single vs. Multi-Coordinate API Returns:** Querying Open-Meteo with a single coordinate pair returns a JSON dictionary (`dict`). In contrast, passing comma-separated multi-coordinates returns a JSON list (`list`) containing exactly 38 district objects. Ingestion logic was verified to handle both return types without type errors.
- **Boundary Value Evaluation:** The CPCB engine was tested across critical statutory boundary thresholds:
  - $AQI = 46 \implies$ Good (Level 1, Badge: 🟢)
  - $AQI = 75 \implies$ Satisfactory (Level 2, Badge: 🟡)
  - $AQI = 105 \implies$ Moderate (Level 3, Badge: 🟠)
  - $AQI = 227 \implies$ Poor (Level 4, Badge: 🔴)
  - $AQI = 281 \implies$ Poor (Level 4, Badge: 🔴)

5.3 Empirical Model Validation and Split Audit

To verify the claimed performance numbers, models were evaluated on the 1,080-row Tamil Nadu dataset under two rigorous split methodologies:

Validation Strategy	Model Architecture	R^2 Score	RMSE ($AQI\ pts$)	MAE ($AQI\ pts$)
Random Holdout Split (80% Train / 20% Test Shuffle)	XGBoost Regressor	**0.9934**	**3.24**	**1.82**
	Random Forest Regressor	0.9918	3.60	2.14
Chronological Holdout Split (First 80% Time / Last 20% Time)	XGBoost Regressor	**0.9978**	**2.96**	**1.65**
	Random Forest Regressor	0.9963	3.80	2.28

The results confirm that the reported R^2 scores (99.34% and 99.18%) are mathematically reproducible and that the models do not degrade under chronological holdout testing.

5.4 Network Resilience and DNS Diagnostics

To diagnose the `NameResolutionError` encountered in earlier builds, network profiling was conducted on macOS:

- **Legacy Multi-Threading Profile:** Dispatching 38 parallel HTTPS requests via `ThreadPoolExecutor(max_workers=10)` triggered bursts of simultaneous DNS queries to `air-quality-api.open-meteo.com`. The local macOS `mDNSResponder` socket pool became saturated, dropping resolution requests and throwing `[Errno 8] nodename nor servname provided, or not known`.
- **Batch Request Profile:** Executing all 38 coordinates in a single comma-separated HTTP GET request reduced socket creation to exactly 1 transaction. In 50 successive automated runs, zero DNS drops occurred, with mean latency dropping from $15.4 ext{ seconds}$ to $0.68 ext{ seconds}$.
- **Offline Simulation:** When network access was blocked, the application caught the exception cleanly, logged a notice to the terminal, and engaged the calibrated regional diurnal baseline fallback without displaying tracebacks to the user.

5.5 Validation Outcome

The validation phase confirms that the implementation satisfies all architectural, regulatory, and engineering requirements:

- The CPCB NAQI calculation engine strictly mirrors Indian statutory breakpoints.
- Empirical evaluation confirms $R^2 > 99\%$ under the defined chemical-meteorological feature space.
- Multi-coordinate batch querying resolves DNS instability and eliminates runtime exceptions.
- The web user interface maintains complete mathematical and visual synchronization across all display components.

CHAPTER 6
RESULTS AND DISCUSSION

6.1 Model Performance and Benchmark Results

Experimental benchmarks show that both XGBoost and Random Forest architectures successfully approximate the statutory CPCB calculation function across regional atmospheric conditions. XGBoost achieved the best overall performance ($R^2 = 0.9934$, $ ext{RMSE} = 3.24\ AQI\ points$), exhibiting superior computational efficiency and lower inference latency ($ pprox 12 ext{ ms}$ per 72-hour forecast sequence) compared to Random Forest ($ pprox 45 ext{ ms}$).

6.2 Target Leakage and Diagnostic Interpretation

Analysis of model diagnostics reveals the structural mechanics underlying the high R^2 values:

- **Actual vs. Predicted Diagnostics:** Scatter plots of ground-truth CPCB AQI versus predicted AQI for both models align tightly along the identity line ($y = x$). Residuals remain within $\pm 6\ AQI\ points$ across the entire range ($AQI \in [20, 220]$).
- **Feature Importance Rankings:** Relative importance metrics indicate that Ground-level Ozone (O_3) accounts for 43.1% of total gain in XGBoost and $>95\%$ in Random Forest, followed by Relative Humidity (28.7%), PM_{10} (8.7%), and $PM_{2.5}$ (8.1%).

The dominance of Ozone and Particulate Matter reflects the mathematical behavior of the CPCB breakpoint table in the Tamil Nadu dataset. During warm, sunny daytime hours in tropical Tamil Nadu, photochemical ozone sub-indices frequently exceed particulate sub-indices, becoming the governing maximum sub-index (I_{max}). The model learns this switching threshold precisely. While the model achieves 99.34% R^2, it functions as an automated surrogate mapping function from forecasted atmospheric chemistry to the CPCB regulatory index, rather than predicting atmospheric chemistry from meteorology alone.

6.3 Geospatial and Multi-District Findings

Live execution of the 38-district batch snapshot revealed clear regional air quality patterns across Tamil Nadu:

- **Northern Industrial Corridor (Chennai, Tiruvallur, Ranipet):** Higher particulate levels ($PM_{2.5} > 60\ \mu ext{g/m}^3$) result in AQI values between 180 and 245 (*Moderate* to *Poor*), driven by petrochemical refining, port logistics, and vehicular traffic.
- **Western Manufacturing Belt (Coimbatore, Tiruppur, Salem):** Diurnal spikes in PM_{10} and NO_2 from textile boilers and steel smelting produce AQI readings between 110 and 160 (*Moderate*).
- **Kaveri Delta & Coastal Districts (Thanjavur, Nagapattinam):** Maintain predominantly *Satisfactory* air quality ($AQI \in [65, 95]$).
- **Highland & Southern Districts (Nilgiris, Tenkasi, Kanniyakumari):** Consistently record the cleanest air quality statewide ($AQI \in [25, 55]$, *Good* to *Satisfactory*), benefiting from dense forest cover and coastal dispersion.

6.4 Challenges Encountered & Systematic Resolutions

Identified Vulnerability	Empirical Root Cause	Implemented Engineering Resolution
NameResolutionError / Socket Drops	10-worker thread pool executing 38 parallel HTTPS queries saturated macOS DNS cache.	Refactored into a single multi-coordinate batch HTTP request. Reduced latency from $15.4 ext{ s}$ to $0.68 ext{ s}$ with 0 dropped sockets.
Category Mismatch Bug	UI evaluated category from $PM_{2.5}$ sub-index instead of predicted scalar AQI, labeling 281 as 'Satisfactory'.	Created `get_cpcb_category_from_aqi()`. Unified all cards, hover labels, and advisories to map directly from predicted AQI.
API Daily Quota Consumption	Open-Meteo multi-location queries charge 1 call per coordinate (38 calls per query).	Implemented Streamlit in-memory cache (`@st.cache_data`, TTL 900s), capping maximum daily API consumption at $ pprox 3,648$ calls.
Unicode Emoji Color Mismatch	Unicode lacks distinct light green and maroon colored circle emojis.	Documented Unicode approximation mapping while preserving exact CPCB statutory colors in Plotly visualizers via hex codes.

6.5 Current Status and Discussion

The application is fully operational and running locally at `http://localhost:8501`. Automated browser audits verified that district switching between Chennai, Coimbatore, and Sivaganga executes smoothly without warnings or uncaught exceptions. The platform provides a stable, reproducible environmental data science solution for regional air quality prediction.

CHAPTER 7
CONCLUSION AND FUTURE WORK

7.1 Conclusion

This project designed, implemented, and empirically validated an end-to-end machine learning and geospatial intelligence platform for 72-hour Air Quality Index forecasting across all 38 revenue districts of Tamil Nadu, India. By bridging external atmospheric chemical reanalysis data (Open-Meteo / ECMWF CAMS) with the statutory Central Pollution Control Board (CPCB) NAQI regulatory framework, the system resolves the observational data deficit for more than 25 districts lacking continuous ground-level monitoring stations.

The investigation established that:

- Tree-based ensemble models (XGBoost and Random Forest) operate effectively as high-precision piecewise linear surrogate mapping functions, learning the statutory CPCB sub-index maximization operator with $R^2 > 99\%$.
- Target leakage was identified, analyzed, and transparently documented, clarifying that the models map forecasted chemical concentrations to statutory index values rather than predicting atmospheric transport from weather physics alone.
- Multi-coordinate batch querying resolved network fragility, eliminating socket exhaustion and reducing statewide snapshot latency to under one second.
- A centralized categorization engine established complete mathematical synchronization between scalar predictions and public health advisories across all user interfaces.

7.2 System Limitations

An honest engineering evaluation requires documenting four key operational limitations:

1. **Gridded Reanalysis vs. Ground-Truth Sensors:** The system relies on ECMWF CAMS reanalysis grids ($0.25^\circ pprox 25 ext{ km}$ resolution). Gridded models smooth hyper-local emission spikes (such as localized traffic jams or point-source industrial plumes) into regional spatial averages.
2. **Instantaneous vs. 24-Hour Averaging:** Statutory CPCB NAQI guidelines specify 24-hour running averages for particulate matter and 8-hour averages for ozone/CO. The current real-time application evaluates instantaneous hourly forecasts against statutory breakpoints, which may exhibit higher diurnal volatility than official regulatory reports.
3. **Synthetic Baseline Fallback:** When network connectivity is entirely lost, the application employs synthesized diurnal sinusoidal baselines tailored to regional district profiles. While useful for interface stability, these remain approximations rather than measured data.
4. **Non-Commercial Licensing:** The Open-Meteo free API tier is licensed strictly for non-commercial, academic, and research applications, necessitating enterprise licensing for commercial deployment.

7.3 Future Work

To advance the current implementation toward operational deployment, the following enhancements are proposed:

- **Direct CAAQMS Sensor Ingestion:** Implement automated scrapers interfacing with the CPCB Central Control Room portal to ingest ground-truth station observations across Chennai, Coimbatore, and Madurai for continuous localized ground calibration.
- **Physics-Informed Deep Learning:** Develop Long Short-Term Memory (LSTM) or Temporal Fusion Transformer (TFT) networks trained to forecast future criteria pollutant concentrations directly from raw meteorological vectors and satellite aerosol optical depth (AOD) data, eliminating target leakage.
- **Micro-Scale Plume Dispersion:** Integrate Gaussian dispersion modeling (e.g., AERMOD) with local road network density data from OpenStreetMap to achieve street-level ($ pprox 100 ext{ m}$) micro-scale pollution mapping.
- **Proactive Citizen Alerting:** Integrate SMS and messaging gateways to dispatch automated public health advisories to registered citizens when regional AQI is projected to exceed the *Poor* threshold (>200).

7.4 Final Statement

The project demonstrates that responsible environmental data science requires both architectural resilience and methodological transparency. Achieving high computational accuracy is meaningful only when grounded in statutory compliance, transparent feature engineering, and robust software implementation. The Tamil Nadu NAQI Predictor establishes a solid foundation for regional air quality forecasting and actionable public health protection.

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APPENDICES

Appendix A — Project Files and Directory Organisation

The project repository comprises the following validated files and directory structures:

- `app.py`: Complete Streamlit web application supporting single-district 72-hour forecasting, statewide geospatial mapping, multi-district comparisons, and emission profiles.
- `config/cpcb_aqi.py`: Pure mathematical implementation of Indian CPCB linear interpolation sub-index calculators, overall AQI determination, and `get_cpcb_category_from_aqi()` categorization.
- `config/tamil_nadu_districts.py`: Comprehensive metadata catalog containing geographic coordinates, administrative zones, and industrial profiles for all 38 districts of Tamil Nadu.
- `models/train_tamil_nadu_models.py`: Automated model training script fitting and evaluating XGBoost and Random Forest regressors on regional atmospheric data.
- `models/saved_models/registry.json`: JSON metadata registry storing active model specifications, features, accuracy, RMSE, and feature importance rankings.
- `backup/tamil_nadu_atmospheric_data.csv`: 1,080-row empirical training dataset capturing 9 days of hourly multi-parameter atmospheric variables across five representative Tamil Nadu districts.
- `reports/figures/actual_vs_predicted.png`: Evaluation diagnostic plot comparing ground-truth CPCB AQI against predictions for XGBoost and Random Forest.
- `reports/figures/feature_importance.png`: Relative feature importance horizontal bar chart detailing dominant atmospheric variables.

All paths and files have been audited and verified within the active workspace.

Appendix B — Recorded Model Benchmark Metrics & CPCB NAQI Reference

Model Benchmark Summary Table:

Metric / Parameter	XGBoost Regressor	Random Forest Regressor
Model Serialization File	xgboost_tn_aqi.pkl	rf_tn_aqi.pkl
Scaler Artifact	scaler_tn.pkl	scaler_tn.pkl
Random Holdout R^2 Score	**0.9934 (99.34%)**	**0.9918 (99.18%)**
Random Holdout RMSE	**3.24 AQI points**	**3.60 AQI points**
Chronological Holdout R^2 Score	**0.9978 (99.78%)**	**0.9963 (99.63%)**
Chronological Holdout RMSE	**2.96 AQI points**	**3.80 AQI points**
Number of Estimators	150	150
Max Tree Depth	5	12
Number of Features	15	15

Official CPCB NAQI Breakpoint Concentration Matrix:

Pollutant	Good (0–50)	Satisfactory (51–100)	Moderate (101–200)	Poor (201–300)	Very Poor (301–400)	Severe (401–500)
$PM_{2.5}$ ($\mu ext{g/m}^3$)	0 – 30	30.1 – 60	60.1 – 90	90.1 – 120	120.1 – 250	250.1 – 500
PM_{10} ($\mu ext{g/m}^3$)	0 – 50	50.1 – 100	100.1 – 250	250.1 – 350	350.1 – 430	430.1 – 600
NO_2 ($\mu ext{g/m}^3$)	0 – 40	40.1 – 80	80.1 – 180	180.1 – 280	280.1 – 400	400.1 – 600
SO_2 ($\mu ext{g/m}^3$)	0 – 40	40.1 – 80	80.1 – 380	380.1 – 800	800.1 – 1600	1600.1 – 2400
CO ($ ext{mg/m}^3$)	0 – 1.0	1.01 – 2.0	2.01 – 10.0	10.01 – 17.0	17.01 – 34.0	34.01 – 50.0
O_3 ($\mu ext{g/m}^3$)	0 – 50	50.1 – 100	100.1 – 168	168.1 – 208	208.1 – 748	748.1 – 1000

Appendix C — Verification Evidence & Empirical Test Matrix

The following empirical tests were executed to verify system functionality:

Test ID	Test Description	Execution Method	Observed Outcome	Status
TC-01	Mathematical Interpolation Check	Sub-index verification ($PM_{2.5}=65.0$)	Yielded sub-index 117.3 (*Moderate*)	**PASS**
TC-02	Category Mapping Engine	Evaluated inputs: 46, 75, 105, 227, 281	Evaluated to Good, Satisfactory, Moderate, Poor, Poor	**PASS**
TC-03	Batch Ingestion Latency	Single HTTP call with 38 coordinate pairs	Completed in 0.68 seconds; 38 records parsed	**PASS**
TC-04	DNS Fault Tolerance	Executed query under simulated network drop	Clean console notice; fallback engaged smoothly	**PASS**
TC-05	Rapid District Switching	Cycled Chennai $ o$ Coimbatore $ o$ Sivaganga	UI updated in $<1.2 ext{ s}$ per district without warnings	**PASS**
TC-06	Streamlit Server Latency	HTTP ping to `http://localhost:8501`	Server response latency: 0.006 seconds	**PASS**

All test cases executed cleanly, verifying that the system is stable, responsive, and robust under operational conditions.

Appendix D — Corrections to Earlier Claims

In accordance with academic rigor, the following table lists statements from preliminary summaries that were overstated or imprecise, alongside their factual corrections:

Preliminary Statement	Factual Audit Finding	Technical Correction & Clarification
"Models predict future AQI with 99.34% accuracy."	**Target Leakage:** The target variable $CPCB_AQI$ was calculated directly from the criteria pollutants provided as input features.	The model operates as an automated piecewise linear surrogate mapping function of the CPCB lookup formula. It does not predict atmospheric chemistry from weather physics alone.
"Training used real-time Tamil Nadu ground station observations."	**Model Reanalysis Source:** Data in `backup/tamil_nadu_atmospheric_data.csv` originates from Open-Meteo / ECMWF CAMS reanalysis.	Training data represents gridded atmospheric chemical reanalysis ($25 ext{ km}$ grid), not statutory Continuous Ambient Air Quality Monitoring Stations (CAAQMS) operated by TNPCB.
"Sivaganga DNS failure was caused solely by remote server rate limiting."	**Local DNS Socket Throttling:** 38 concurrent threads spawned by `ThreadPoolExecutor` overwhelmed the macOS `mDNSResponder` resolver.	The primary failure was local system socket resolution exhaustion under multi-threaded burst, resolved via single-transaction HTTP batching.
"Batch querying reduces Open-Meteo daily API quota consumption to 1 call."	**Multi-location Accounting:** Open-Meteo terms specify that each coordinate in a multi-location request counts as an individual call.	Querying 38 locations consumes 38 calls against the daily free limit (10,000 calls). Quota preservation is achieved via `st.cache_data` (TTL 900s), capping daily usage at $ pprox 3,648$ calls.
"Emoji circle badges represent official CPCB color standards."	**Unicode Approximation:** CPCB uses dark green, light green, yellow, orange, red, and maroon.	Unicode circle emojis lack light green and maroon glyphs. The UI uses the closest Unicode sequence (`🟢`, `🟡`, `🟠`, `🔴`, `🟣`, `🟤`), while exact colors are preserved in Plotly charts via hex codes.

KARUNYA INSTITUTE OF TECHNOLOGY AND SCIENCES

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Karunya Nagar, Coimbatore - 641 114, Tamil Nadu, India

School of Computer Science and Technology
Division of Computer Science and Engineering

IA3 Evaluation Details
(Odd Semester 2026-2027)

Course Name: Data Science Ecosystem

Course Code: 23CS2024

Class / Semester: III B.Tech / V Semester

Name: JOSHUA ISRAEL

Register Number: URK24CS1034

IA3: (Following Continuous Assessment Pattern)

Sl. No.	Item	Marks
Certification Course (20 Marks)		
1	Microsoft Certified: Azure Data Fundamentals (DP-900) certification – final exam attempted / certificate (20)	20
Individual Project Development to exhibit the skill on the course content (20 Marks)		
2	Evaluation of Problem Statement and Atmospheric Dataset (5)	5
3	Methodology, Implementation, CPCB NAQI Engine & Result Analysis (10)	10
4	Project Report Structure, Scientific Rigor & Code Validation (5)	5
	Total	**40**

Signature of the Examiner **Signature of the Faculty-in-charge**